

AI Ethics - Team15

SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models

Manakul, P., Liusie, A., & Gales, M. J. F. (2023). SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. arXiv.

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Agenda

1 Problem

2 Related work

3 Replication approach

4 Improvement approach

5 Data

6 Plan



1. Problem

SELFCKEKGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models

This paper focuses on hallucination detection in LLMs outputs, especially in black-box settings where the model only returns text and does not expose token probabilities or internal states. Large Language Models (LLMs) such as GPT-3 are capable of generating fluent and convincing text, but they frequently produce hallucinations, statements that are false, fabricated, or not grounded in real-world facts.

These hallucinations are problematic because they are presented with high confidence, making them difficult for users to detect.

RESEARCH QUESTION

How can we detect if a statement generated by an LLM is factual or hallucinated, using only its output and without relying on internal model information or external data?



Existing Approaches for Detecting Hallucination

1. Fact Verification

- Fact checking w/ external database
- A survey on automated fact-checking. (Guo et al., 2022)

2. Grey-Box method

- Uncertainty-based Assessment
- Requires output token-level probability
- Entropy

3. Black-Box method

- Proxy LLMs



2. Related work

Current limitations in fact-checking approaches



Access to Output Probability Distribution

Uncertainty metrics such as token probability or entropy require access to token-level probability distributions, information which may not be available to users for example when systems are accessed through limited external APIs (e.g. systems such as ChatGPT).



Access to External Databases

Facts can only be assessed relative to the knowledge present in the database. Also, some databases are interfaced via separate, often complex, modules.



Cost

Existing fact-verification approaches follow a multi-stage pipeline of claim detection, evidence retrieval and verdict prediction (Guo et al., 2022; Zhong et al., 2020). Such methods, however, require access to external databases and can have considerable inference costs.



3. Replication Approach

SelfCheck GPT

Zero-resource Black-box hallucination detection

Core Idea

Factual → Sampled responses are consistent

Hallucinated → Sampled responses diverged

- Black-box: No internal access needed
- Zero-resource: No external DB required
- Outperforms grey-box baselines

Key Insight

Known facts yield consistent responses;
Hallucinated facts diverge — no external DB needed.



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How it works

1. Generate

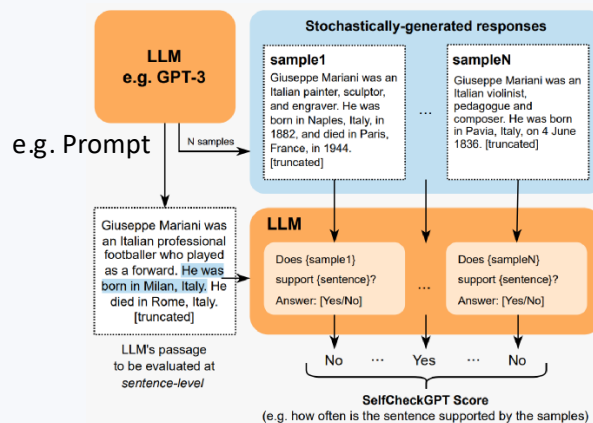
Response R (temp 0)

2. Sample

N Responses (temp 1)

3. Compare

Consistent?



5 variants: BERTScore, QA, N-gram, NLI, Prompt

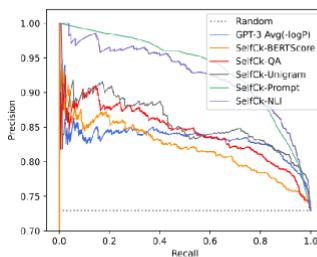


3. Replication Approach - Evaluation

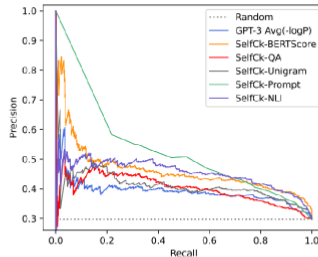
1. Compute Hallucination Scores
2. Sentence-Level Evaluation
 - Measure detection accuracy using AUC-PR against human GT.
3. Passage-Level Evaluation
 - Assess overall text factuality using Pearson/Spearman Correlation.

**SelfCheck-NLI and SelfCheck-prompt
achieve best performance**

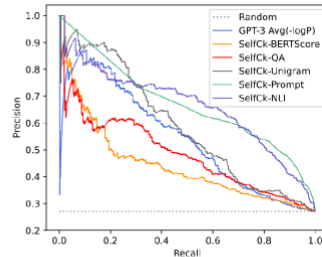
Sentence Level



(a) Non-Factual Sentences

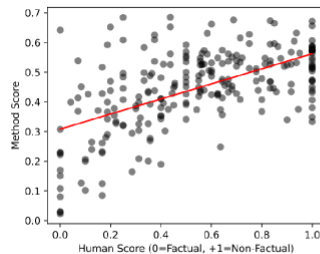


(b) Non-Factual* Sentences

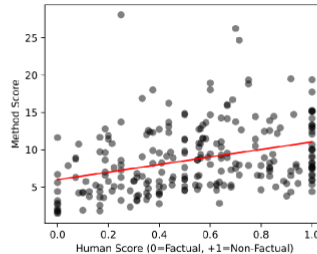


(c) Factual Sentences

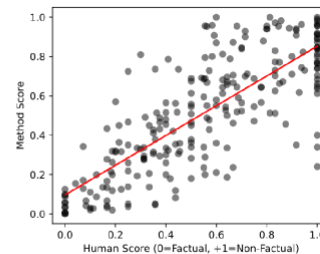
Passage Level



(a) GPT-3 Avg(-log p)



(b) LLaMA-30B Avg($\mathcal{H}$)



(c) SelfCheckGPT-Prompt



3. Replication Approach - Evaluation

- Results in **Figure 7** show that the performance of SelfCheckGPT increases smoothly as more samples are used, with diminishing gains as more samples are generated.
- Since the n-gram variant requires more samples to stabilize, we decided not to focus on it for replication.

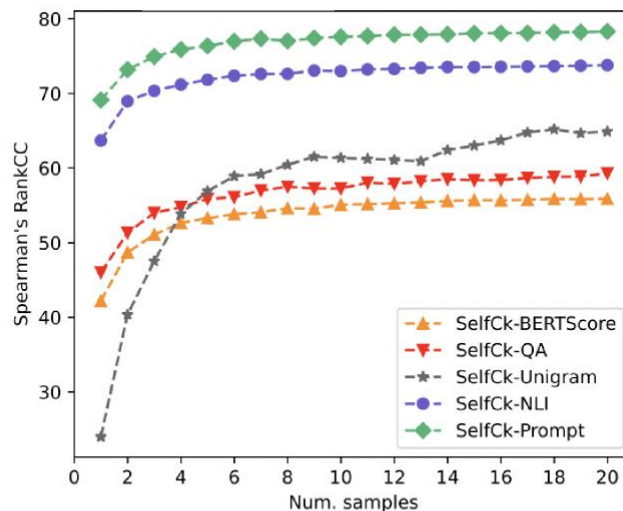


Figure 7: The performance of SelfCheckGPT methods on ranking passages (Spearman's) versus the number of samples.



4. Improvement approach

Improvement 1

Find out and analyze the hallucination that cannot be detected by SelfCheckGPT
-- that is, the **hallucination with high certainty**

Q. Does it exist?

Q. If then, how much?

Improvement 2

Use **math or coding task** which are not natural language but scientific terms

Why?

Recent expansion of LLMs' deployment for coding, reasoning, and scientific tasks

Easy to implement and evaluate the system's performance without labor-intensive verification



5. Data Overview

GSM8K Dataset

Cobbe et al., 2021

A large-scale dataset of grade school 8,500 **math problems**.

Each entry contains: A problem **question** and the step-by-step **solution**

A family of 6 (2 adults and 4 kids) are to divide a watermelon such that each adult gets a slice that is twice as big as that of each kid. What percentage of the watermelon does each adult get?

[Step 1- Assumptions]

1. The watermelon is divided into equal parts for the kids and twice as large parts for the adults.
2. There are 2 adults and 4 kids in the family.
3. The total watermelon is considered as 100%.

[Step 2] First, we need to find the total number of parts the watermelon is divided into. Each adult gets 2 parts and each kid gets 1 part.



[Solution] Each adult gets 25% of the watermelon.

Question

Step-by-step solution

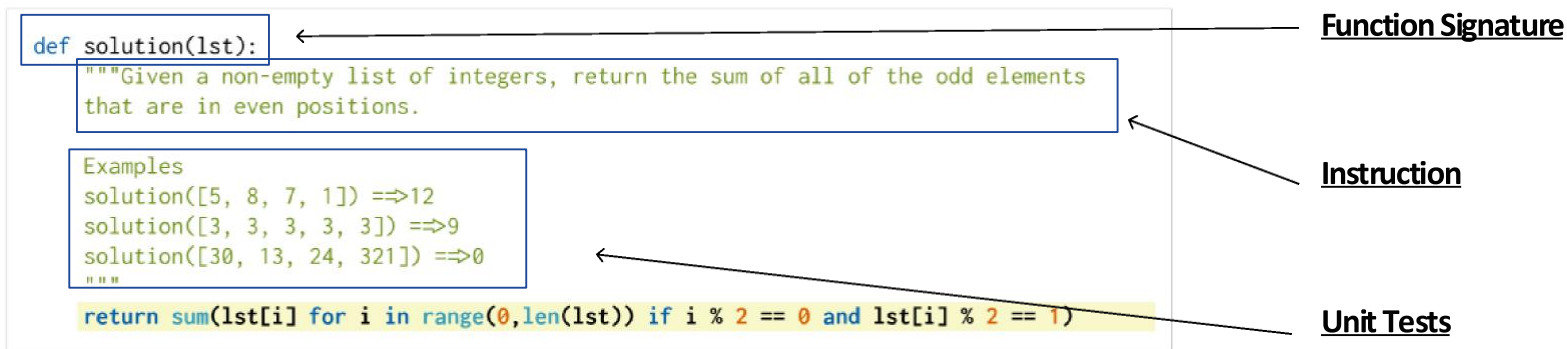


5. Data Overview

HumanEval Dataset

Chen et al., 2021

A dataset of 164 hand-written Python programming problems.
Each entry contains: a **function signature**, **instruction** and **unit tests**



5. Planned dataset

New Datasets - GSM8K, HumanEval

Dataset	Domain	# Topics	# Passages	Annotation
WikiBio (original)	Biography	238	238 + 20 x 238	Factuality (Major Inaccurate, Minor Inaccurate, Accurate) per sentence
New - GSM8K (Math)	Grade School Math	~200	200 + 20 x 200	No need → Correctness
New - HumanEval (Coding)	Python Coding	~164	164+ 20 x 164	No need → Correctness

SelfCheckGPT Variants

GSM8K (Math): NLI, Prompt, and BERTScore

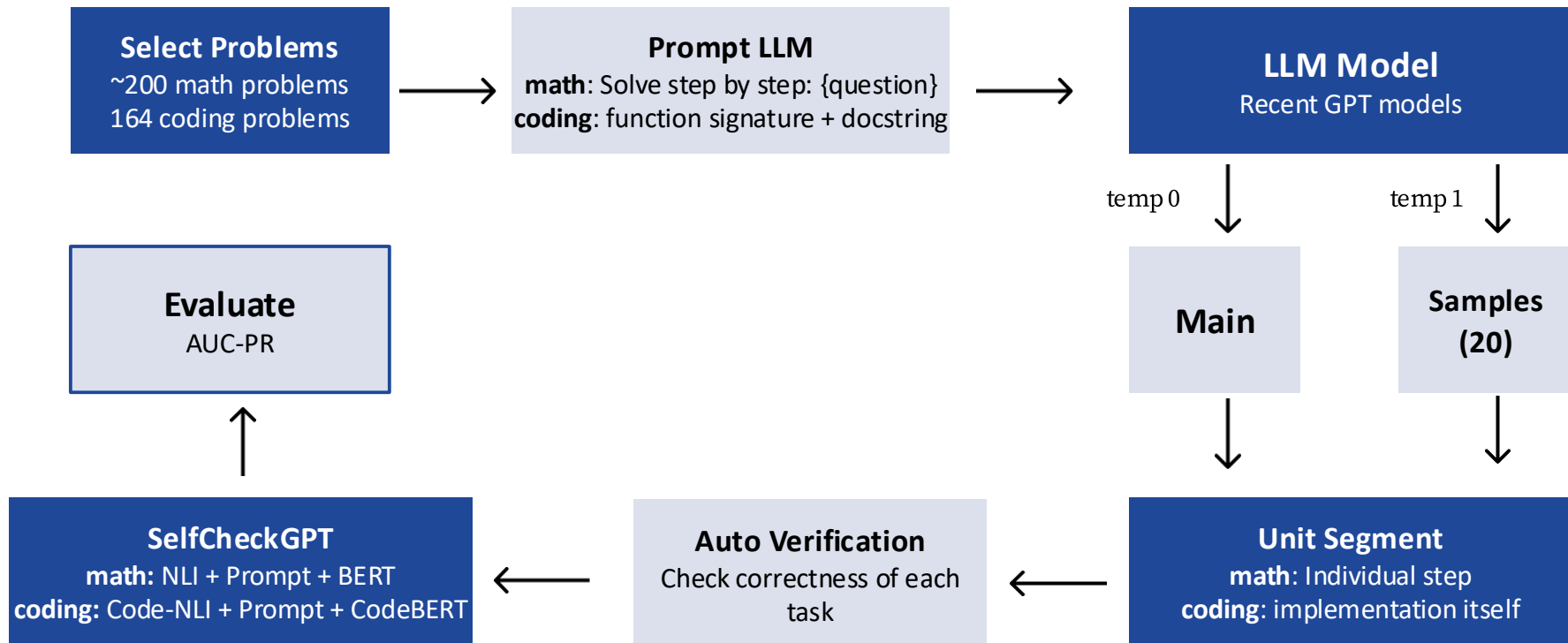
- **N-gram** skipped: the semantic difference between math expression and language tokens
- **QA** skipped: cannot produce meaningful choices from math expressions

HumanEval (Coding): Code-NLI, Prompt, and CodeBERTScore

- **N-gram** and **QA** are skipped due to the similar reason as Math's
- **Code-NLI** and **CodeBERT** to use backbone network trained with code snippets



5. Data Pipeline



6. Plan



Thank you!

